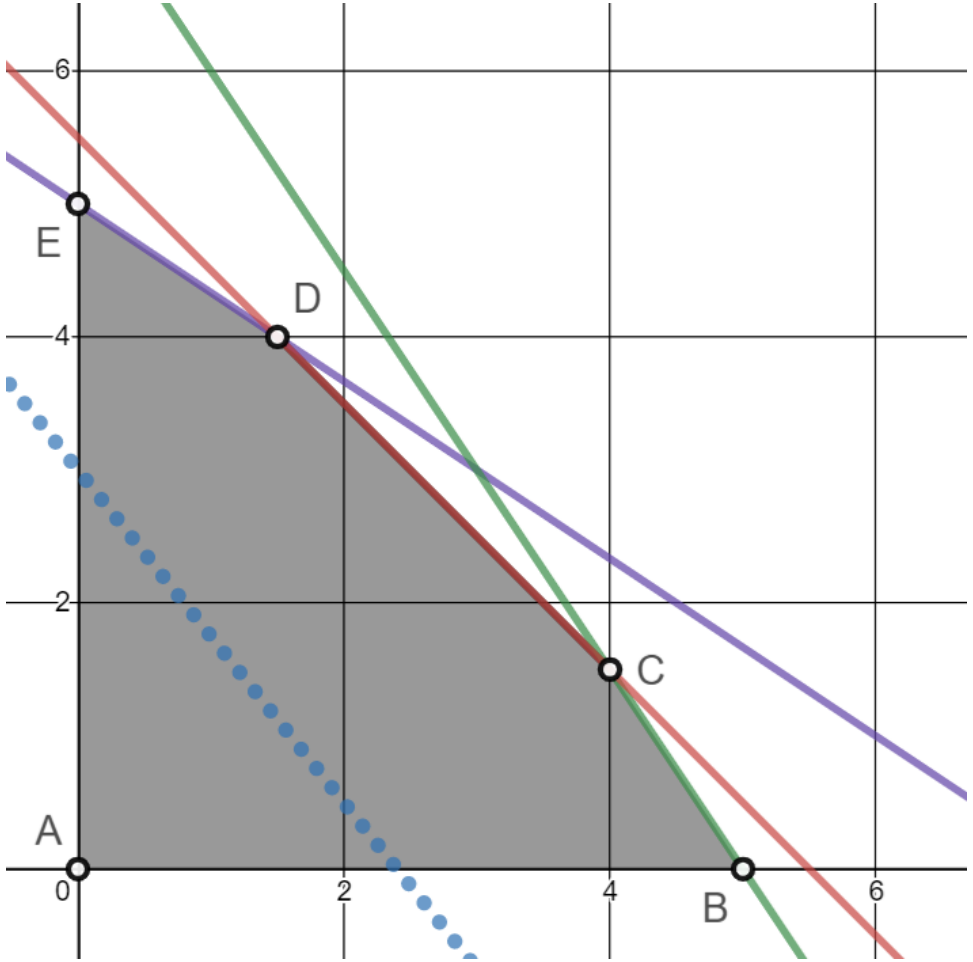
Graphical Demonstration of Slack Variables and binding/non-binding constraints

Standard Linear Program



maximise

$$P = 5x + 4y$$

subject to:

$$2x + 3y \leq 15$$
$$3x + 2y \leq 15$$
$$2x + 2y \leq 11$$

and

$$x, y \geq 0$$

Point A has an objective value of

$$P = 5 \cdot 0 + 4 \cdot 0 = 0$$

Introduce slack variables

maximise

$$P = 5x + 4y$$

subject to:

$$2x + 3y \leq 15$$
$$3x + 2y \leq 15$$
$$2x + 2y \leq 11$$

and

$$x, y \geq 0$$



maximise

$$P = 5x + 4y$$

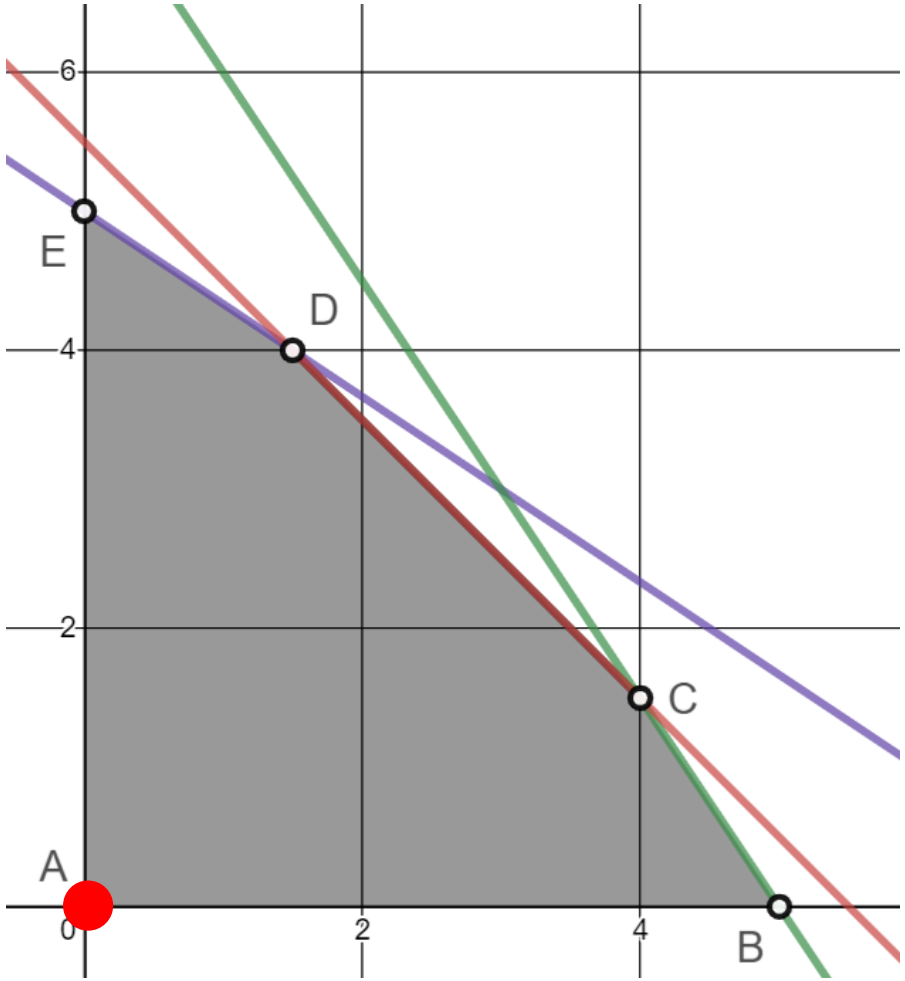
subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

and

$$x, y, s_1, s_2, s_3 \geq 0$$

Note: This is called Standard Form



maximise

$$P = 5x + 4y$$

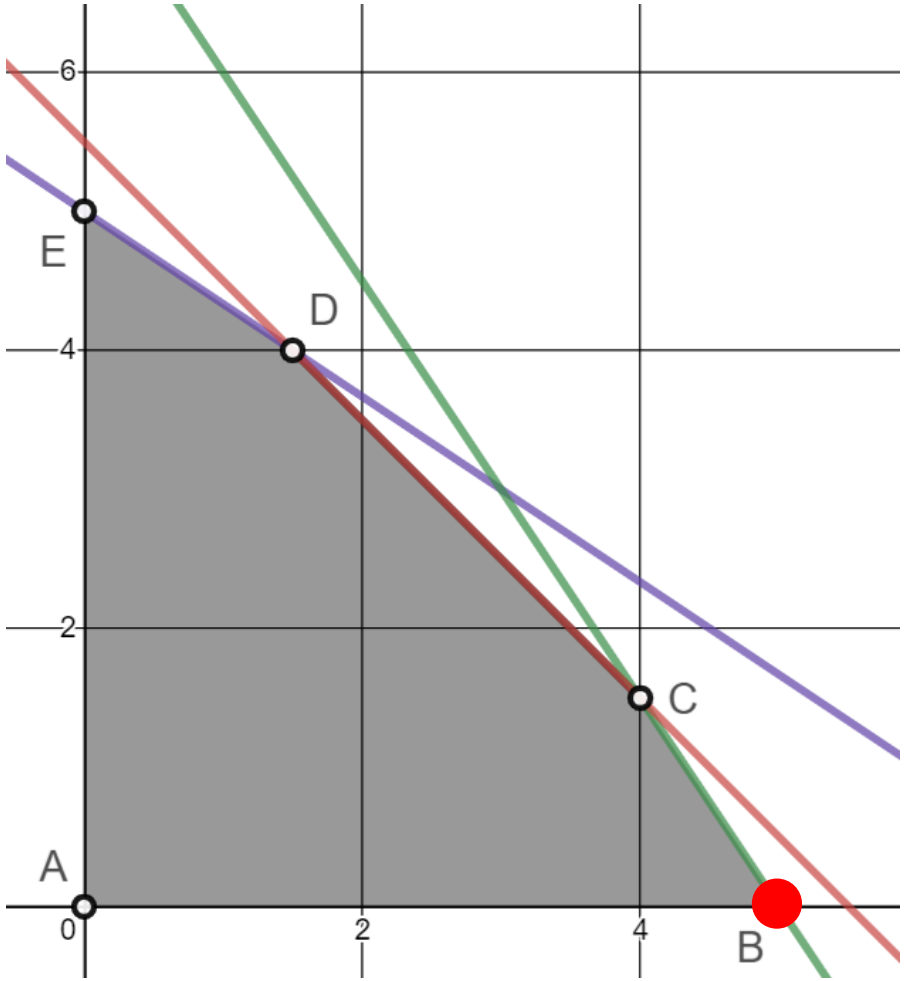
subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

and

$$x, y, s_1, s_2, s_3 \geq 0$$

	x	y	s_1	s_2	s_3	P
A	0	0	15	15	11	0
B						
C						
D						
E						



maximise

$$P = 5x + 4y$$

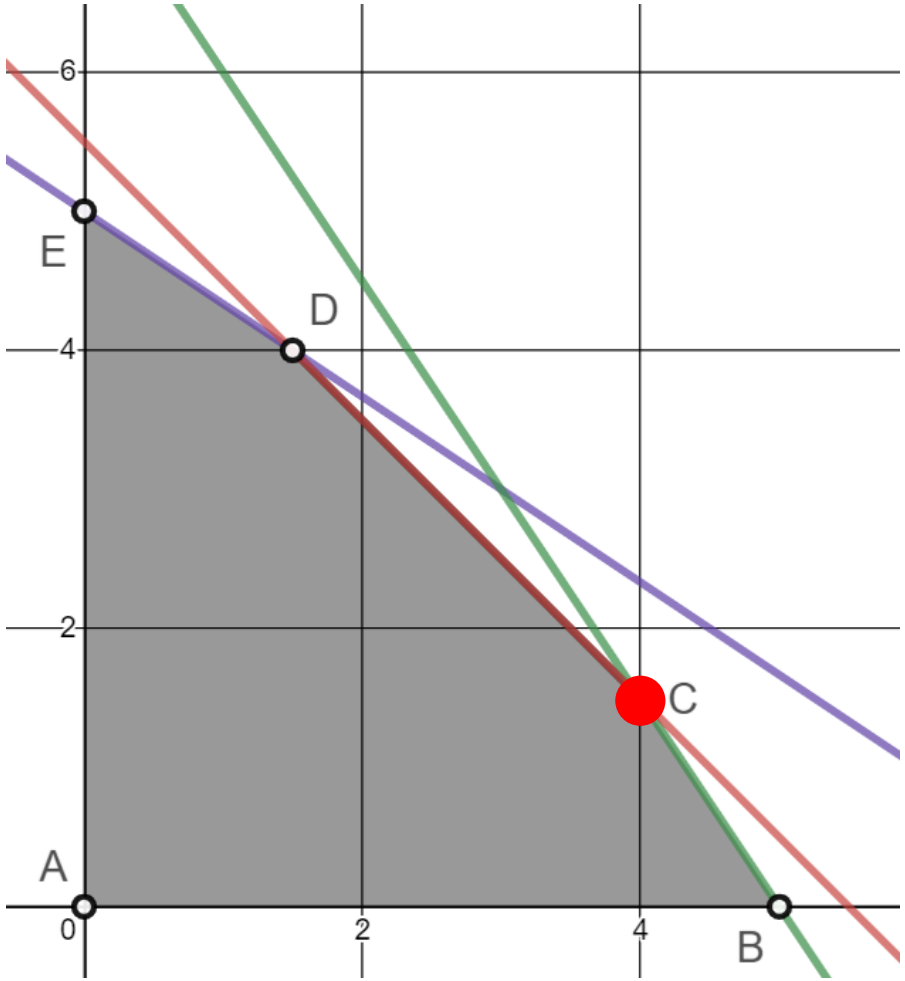
subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

and

$$x, y, s_1, s_2, s_3 \geq 0$$

	x	y	s_1	s_2	s_3	P
A	0	0	15	15	11	0
B	5	0	5	0	1	25
C						
D						
E						



maximise

$$P = 5x + 4y$$

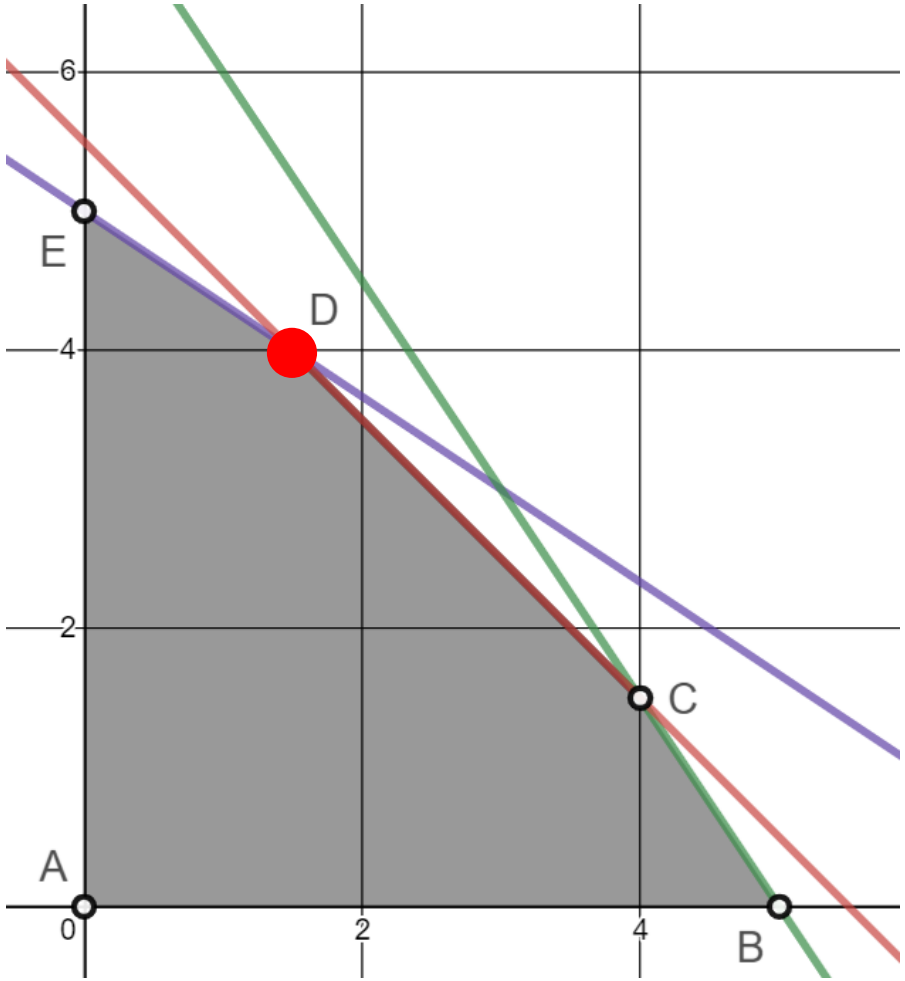
subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

and

$$x, y, s_1, s_2, s_3 \geq 0$$

	x	y	s_1	s_2	s_3	P
A	0	0	15	15	11	0
B	5	0	5	0	1	25
C	4	1.5	2.5	0	0	26
D						
E						



maximise

$$P = 5x + 4y$$

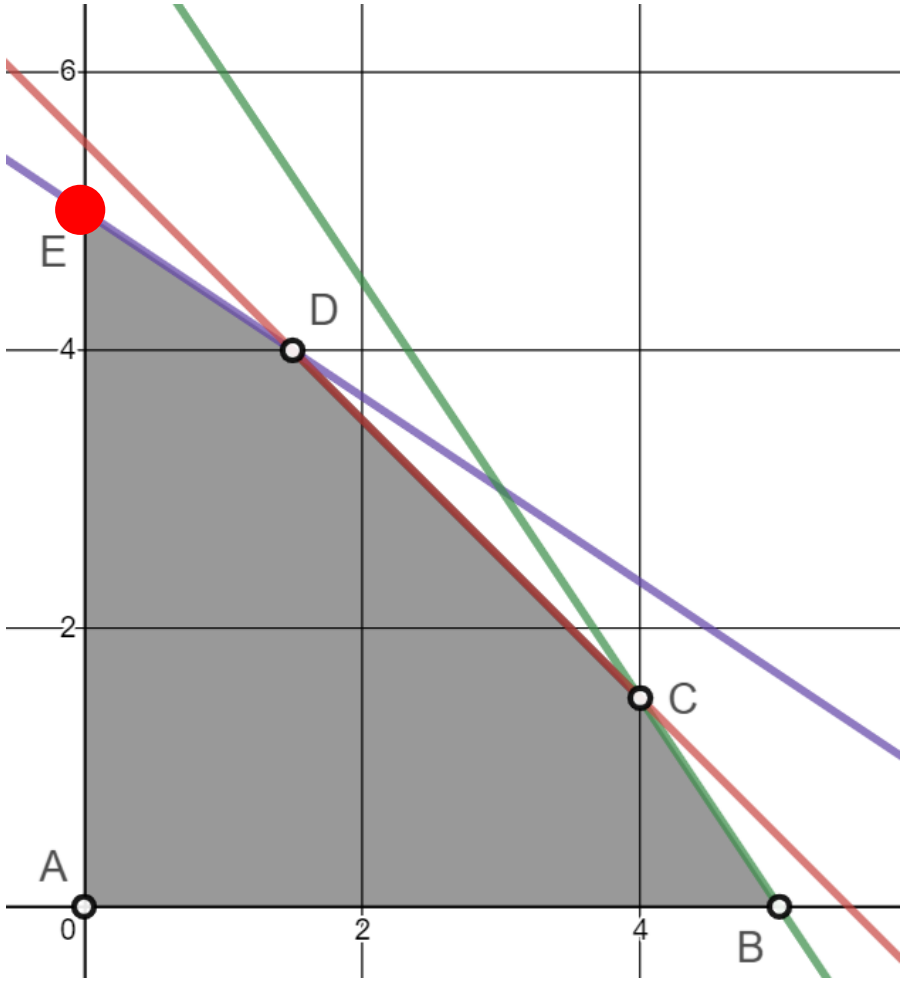
subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

and

$$x, y, s_1, s_2, s_3 \geq 0$$

	x	y	s_1	s_2	s_3	P
A	0	0	15	15	11	0
B	5	0	5	0	1	25
C	4	1.5	2.5	0	0	26
D	1.5	4	0	2.5	0	23.5
E						



maximise

$$P = 5x + 4y$$

subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

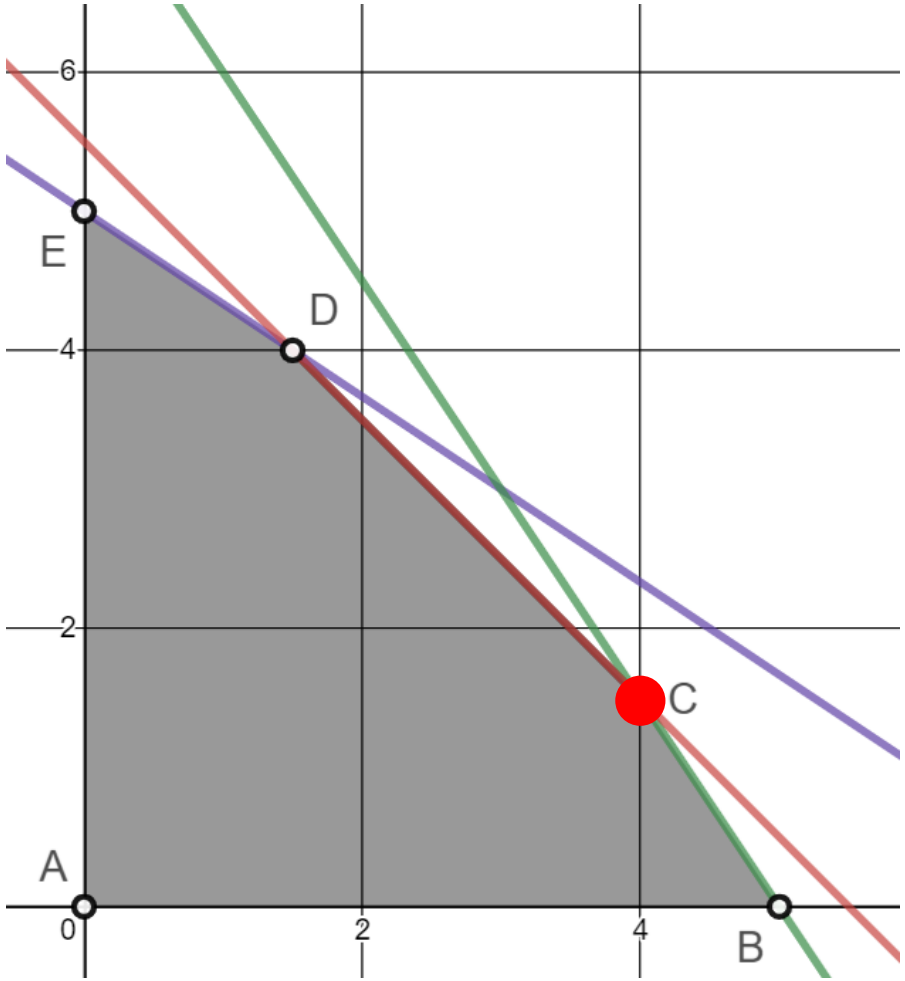
and

$$x, y, s_1, s_2, s_3 \geq 0$$

	x	y	s_1	s_2	s_3	P
A	0	0	15	15	11	0
B	5	0	5	0	1	25
C	4	1.5	2.5	0	0	26
D	1.5	4	0	2.5	0	23.5
E	0	5	0	5	1	20

Therefore C is optimal at $(x, y)^* = (4,1.5), P = 26$

Notice that here $s_1 = 2.5, s_2 = 0, s_3 = 0$. Hence the purple constraint is non-binding, but the green and red are!



maximise

$$P = 5x + 4y$$

subject to:

$$2x + 3y + s_1 = 15$$
$$3x + 2y + s_2 = 15$$
$$2x + 2y + s_3 = 11$$

and

$$x, y, s_1, s_2, s_3 \geq 0$$

	x	y	s_1	s_2	s_3	P	
A	0	0	15	15	11	0	
B	5	0	5	0	1	25	
C	4	1.5	2.5	0	0	26	*
D	1.5	4	0	2.5	0	23.5	
E	0	5	0	5	1	20	

Slack Variable ‘vs’ Surplus Variable

- Slack variables measure the amount remaining. e.g. unused manufacturing hours.
- Used for ≤ (less than or equal to) constraints

$$2x + 3y \leq 15$$

Becomes

$$2x + 3y + s_1 = 15$$

- Surplus variables measure the amount exceeded. e.g. over production
- Used for ≥ (greater than or equal to) constraints

$$2x + 3y \geq 15$$

Becomes

$$2x + 3y - s_1 = 15$$